Hilbert Metric

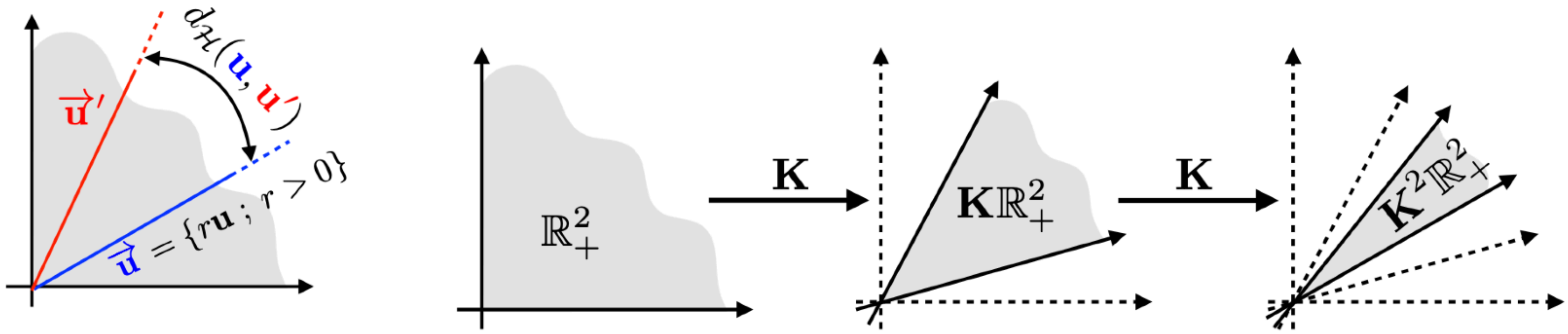
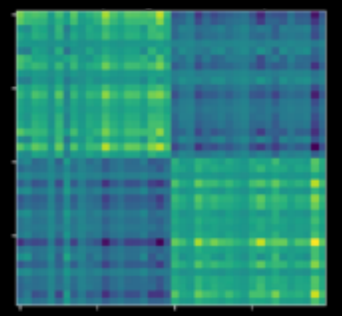


Figure 4.7: Left: the Hilbert metric $d_{\mathcal{H}}$ is a distance over rays in cones (here positive vectors). Right: visualization of the contraction induced by the iteration of a positive matrix $\mathbf{K}$.

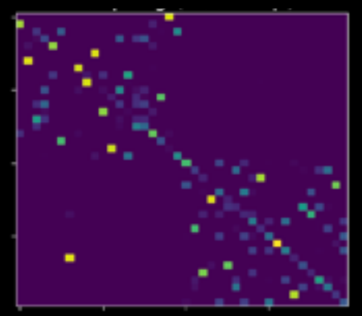
Figure from Peyre & Cuturi (2009)

Trade-offs in Choosing ε



Large ε:

- Coupling is smoother ("blurrier" transport).
- Converges fast, numerically stable.
- But solution may be too far from true OT.



Small ε:

- Coupling π_ε close to *true* OT solution.
- Slow convergence, more iterations needed.
- Numerical issues: under/overflow in $\mathbf{K} = e^{-\mathbf{C}/\varepsilon}$.